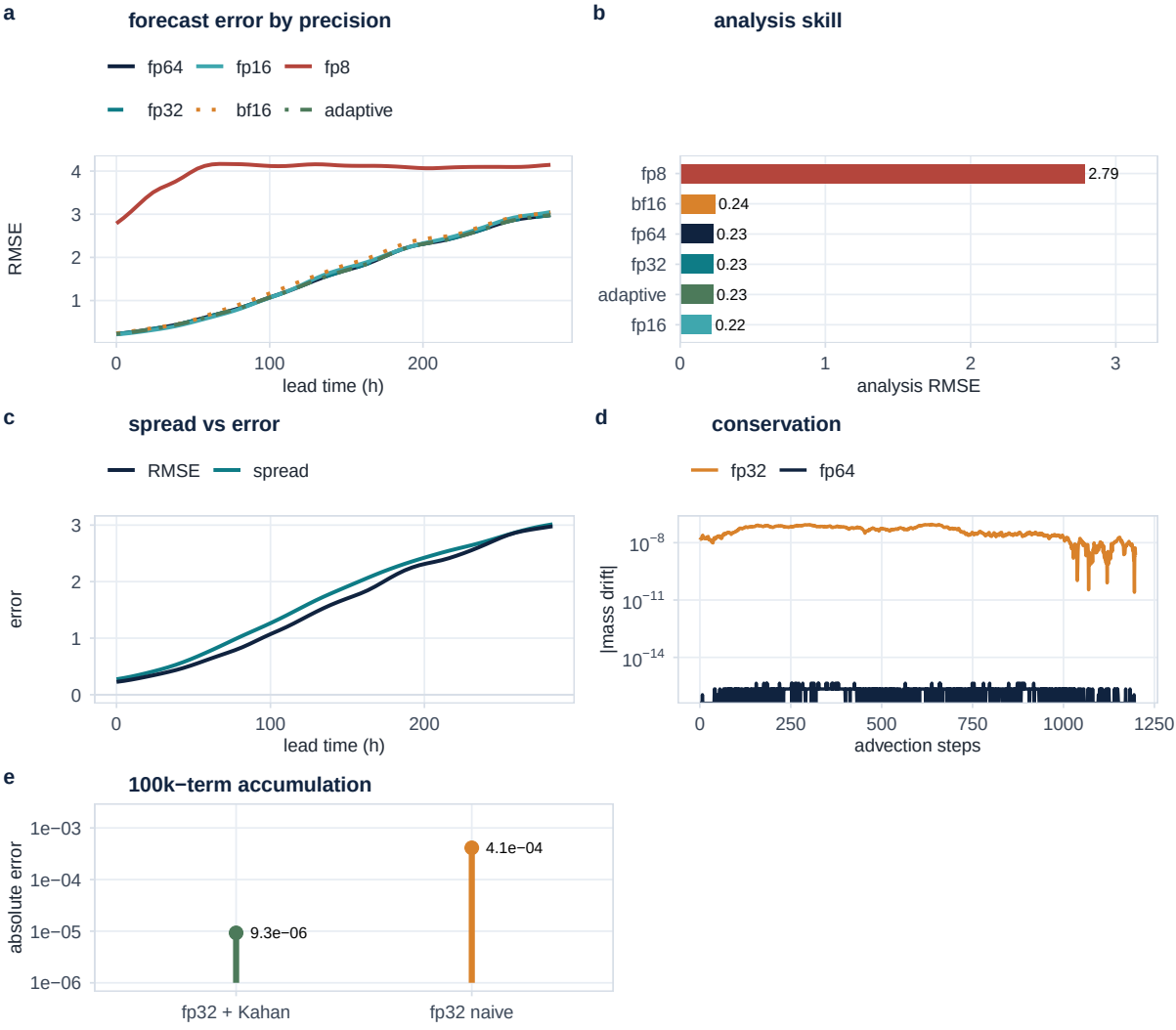
a
forecast error by precision
fp64
fp16
fp8
fp32
bf16
adaptive
RMSE
4
3
2
1
0
100
200
lead time (h)
b
analysis skill
fp8
bf16
fp64
fp32
adaptive
fp16
2.79
0.24
0.23
0.23
0.23
0.22
0
1
2
3
analysis RMSE
c
spread vs error
RMSE
spread
error
3
2
1
0
0
100
200
lead time (h)
d
conservation
fp32
fp64
|mass drift|
10−8
10−11
10−14
0
250
500
750
1000
1250
advection steps
e
100k−term accumulation
absolute error
1e−03
1e−04
1e−05
1e−06
4.1e−04
9.3e−06
fp32 + Kahan
fp32 naive